Brac University
Department of Computer Science and Engineering

CSE111: Programming Language II   Summer 2024   Quiz 2   Marks: 10

Name:	ID:	Section:

1. **Create/Design** a class called **Laptop** such that the results displayed in the output box are generated. [You are not allowed to change the code below]

[10]

Driver Codes/Tester Class:

```
public class LaptopTester {
   public static void main(String[] args) {
      Laptop laptop1 = new Laptop();
      laptop1.printDetails();
      laptop1.model = "Dell Inspiron";
      System.out.println("1###################");
      laptop1.installApp("Chrome");
      laptop1.printDetails();
      System.out.println("2###################");
      laptop1.installApp("VS Code");
      laptop1.installApp("Slack");
      laptop1.printDetails();
      System.out.println("3###################");
      laptop1.installApp("Zoom");
      laptop1.printDetails();
      System.out.println("4###################");
      laptop1.uninstallApp("VS Code");
      laptop1.printDetails();
      System.out.println("5###################");
      laptop1.searchApp("Slack");
      laptop1.searchApp("Spotify");
   }
}
```

Outputs:

```
Laptop Model unknown
Applications Installed 0
1###################
Application Installed
Laptop Model Dell Inspiron
Applications Installed 1
Installed Applications:
Chrome
2###################
Application Installed
Application Installed
Laptop Model Dell Inspiron
Applications Installed 3
Installed Applications:
Chrome
VS Code
Slack
3###################
Memory full. New application can't be installed.
Laptop Model Dell Inspiron
Applications Installed 3
Installed Applications:
Chrome
VS Code
Slack
4###################
Application Uninstalled
Laptop Model Dell Inspiron
Applications Installed 2
Installed Applications:
Chrome
Slack
5###################
Application Slack is installed.
Application Spotify is not installed.
```

Solution:

```
public class Laptop {
  String model = "unknown";
  String[] apps = new String[3];
  int appCount = 0;

  public void printDetails() {
    System.out.println("Laptop Model " + model);
    System.out.println("Applications Installed " + appCount);
    if (appCount > 0) {
      System.out.println("Installed Applications:");
      for (int i = 0; i < appCount; i++) {
        System.out.println(apps[i]);
      }
    }
  }

  public void installApp(String appName) {
    if (appCount < apps.length) {
      apps[appCount] = appName;
      appCount++;
      System.out.println("Application Installed");
    } else {
      System.out.println("Memory full. New application can't be installed.");
    }
  }

  public void uninstallApp(String appName) {
    int index = -1;
    for (int i = 0; i < appCount; i++) {
      if (apps[i].equals(appName)) {
        index = i;
        break;
      }
    }
    if (index != -1) {
      for (int i = index; i < appCount - 1; i++) {
        apps[i] = apps[i + 1];
      }
      apps[appCount - 1] = null;
      appCount--;
      System.out.println("Application Uninstalled");
    } else {
```

```java
            System.out.println("Application not found");
        }
    }

    public void searchApp(String appName) {
        boolean found = false;
        for (int i = 0; i < appCount; i++) {
            if (apps[i].equals(appName)) {
                found = true;
                break;
            }
        }
        if (found) {
            System.out.println("Application " + appName + " is installed.");
        } else {
            System.out.println("Application " + appName + " is not installed.");
        }
    }
}
```